

Masked IRL Reproduction

Scope. I reproduced the Masked IRL learning pipeline as closely as the released material allowed. The public repository did not include the simulation trajectories and resource/URDF package used for the paper's main simulation, so the results below use a fixed replacement benchmark. I therefore treat this as a controlled, partially paper-faithful reconstruction rather than an exact numerical reproduction.

1. What I implemented

I started from the released code and kept the original source behavior separate from the paper-aligned version. The final V3 setup uses the same fixed trajectories for every method and seed, so the comparison reflects the learning protocol rather than a changing dataset.

Item	Released code	V3 used here
T5 encoder	Trainable	Frozen
Mask perturbation	Gaussian / joint	Uniform(0,1) / local
Mask weight	1	10
Training schedule	Different / incomplete match	1000 pretrain + 100 adapt
Validation	Held-out identities reused	Training identities only

2. Main experience with the code

- The largest reproducibility issue was missing simulation data and environment resources. I reconstructed a controlled replacement benchmark and kept that limitation explicit.
- The paper and released implementation differ in several important places, especially T5 freezing, the masking loss, perturbation noise, lambda, and the training schedule.
- I separated paper-style few-shot adaptation from a strict zero-shot test. This avoids calling a preference "unseen" after it has already supplied demonstrations or gradients.
- For the local masking loss, I used chunked computation only after checking that the chunked and unchunked losses and gradients were numerically equivalent.

3. Baseline results

I compared LC-RL, Explicit Mask, and Masked IRL with oracle masks using three prespecified seeds (12345, 23456, 34567). The metric is pairwise trajectory-ordering accuracy within matched start-goal groups; it is not robot task-success rate.

Method	Before adaptation	Few-shot adapted	Strict zero-shot
LC-RL	62.93 ± 1.44	65.07 ± 0.81	62.09 ± 1.09
Explicit Mask	67.66 ± 3.15	70.38 ± 1.48	65.44 ± 2.21
Masked IRL	67.97 ± 2.02	69.41 ± 1.02	66.26 ± 1.70

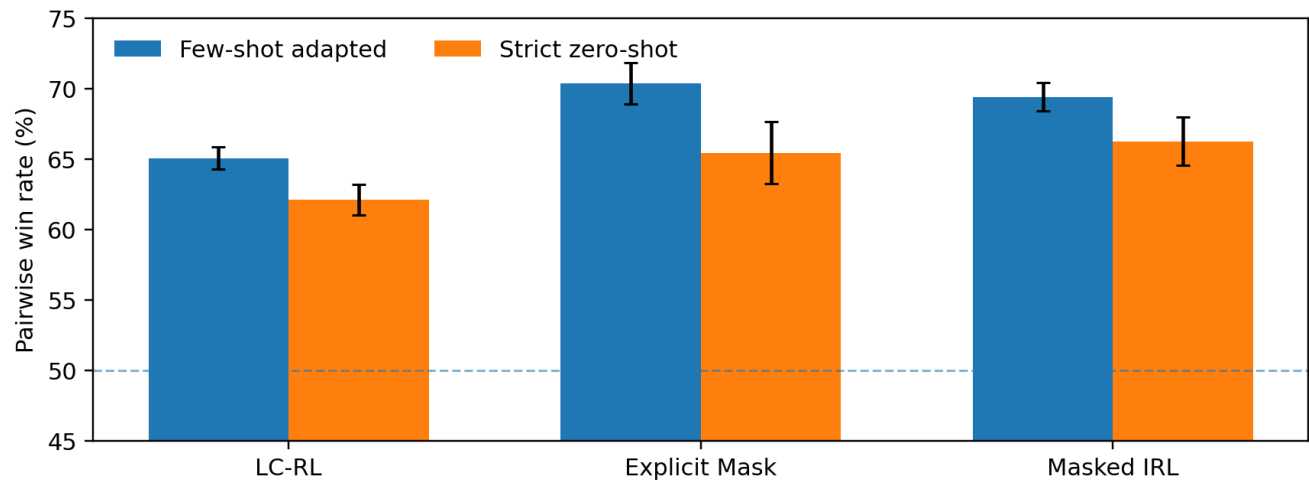


Figure 1. Core comparison. Error bars are standard error across three training seeds.

The result is mixed rather than one-sided. Explicit Mask has the highest few-shot adapted mean (70.38%), while Masked IRL has the highest strict zero-shot mean (66.26%). Both mask-based methods are above LC-RL in this benchmark, but the three-seed results do not justify a universal superiority claim.

I also evaluated the author-saved LLM masks instead of regenerating masks with a new API. Performance dropped to 58.33% for Explicit Mask and 59.43% for Masked IRL after adaptation. This was the clearest indication that mask quality is a practical bottleneck, not just a preprocessing detail.

4. What the baseline does and does not show

- It shows that the released learning components can be reconstructed into a consistent multi-seed comparison on fixed data.
- It does not reproduce the paper's original numerical values because the original simulation trajectories and assets are unavailable.
- Three seeds are enough to see the main trends, but they are not enough for broad statistical claims or claims across new environments.

5. Limitation I tested: incorrect relevance masks

The method assumes that the relevance mask correctly identifies which state variables matter. I tested this directly by flipping mask bits at increasing rates and then retraining/evaluating the reward learner. The same corrupted masks were used for matched methods and seeds.

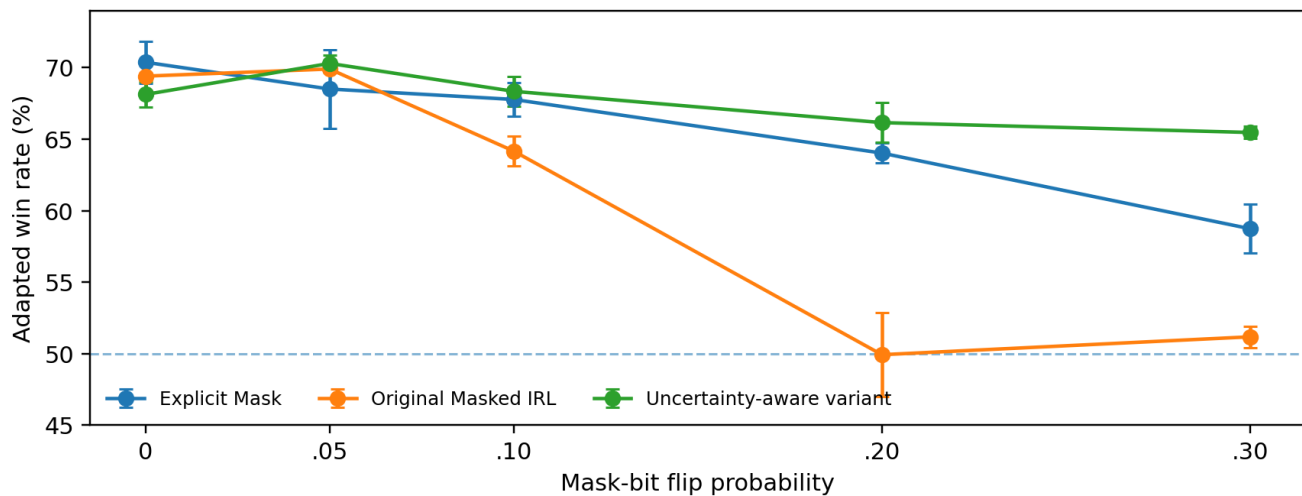


Figure 2. Few-shot adapted performance under mask corruption. The uncertainty-aware variant is the exploratory modification described below.

The unexpected result was that original Masked IRL was not automatically more robust. At 30% corruption, adapted accuracy fell to 51.18%, compared with 58.74% for Explicit Mask. This is a useful negative finding: the invariance penalty can become harmful when an actually important coordinate is labelled irrelevant.

6. Proposed change

I therefore tested a simple uncertainty-aware version of the masking loss. Instead of treating every zero in the mask as equally trustworthy, the regularization is weakened when the available demonstrations show that a supposedly irrelevant coordinate has a noticeably different state distribution. The model still uses the original mask, but it does not enforce every irrelevance decision with the same strength.

At 30% mask corruption: the proposed variant reached $65.47 \pm 0.42\%$ adapted accuracy, compared with $51.18 \pm 0.75\%$ for original Masked IRL and $58.74 \pm 1.71\%$ for Explicit Mask. That is +14.29 percentage points over original Masked IRL in the paired three-seed comparison.

I would treat this as a promising direction, not a finished new method. It performs slightly worse in the clean condition, it was designed after seeing the baseline failure, and part of the gain may come from reducing the total regularization strength. The next clean test would be a fresh benchmark with mass-matched controls and five prespecified seeds.

7. Final takeaway

The main outcome of this task was not an exact reproduction number. It was a reproducible baseline that exposed where the public release, the paper description, and the method assumptions differ. The strongest practical limitation I found is sensitivity to imperfect relevance masks. The uncertainty-aware result suggests that making the invariance penalty less rigid is worth studying further, but it still needs independent confirmation.

Code and reproducibility files: provided in the accompanying GitHub repository, including run configs, result tables, verification scripts, and the fixed replacement benchmark.

Reference: Masked IRL: LLM-Guided Reward Disambiguation from Demonstrations and Language, arXiv:2511.14565; MIT-CLEAR-Lab/Masked-IRL